The Oracle Gap Has a Floor: Separating Aleatoric and Deployable Headroom in Evaluating Label Acquisition

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Abstract. Active-learning studies bound achievable performance with oracles that select labels in hindsight, and report the oracle's lead over practical strategies as headroom. We show that, when outcomes are noisy, much of this lead is unreachable. We split the oracle gap into test-set reuse, an aleatoric floor created by the randomness of the acquired labels, and a deployable gap, and estimate all three on real data with a probability model fitted on disjoint records. Across eight credit, readmission, marketing, e-mail and digit cohorts and three learners (logistic, boosting and a neural network), the floor is material in nearly every cohort and its share grows with label noise: on the raw-pixel digit task used by a recent benchmark the oracle's lead is mostly real, whereas on noisy credit and readmission outcomes 58–100% of it is test reuse and floor. A closed-form bracket built from near-tied leading sets contains the measured floor in all 87 settings we examined, and under a risk budget the floor rises with the budget in every cohort. Which strategy closes the remaining deployable gap depends on whether its information proxy matches the learner; a risk-conservative policy, DC-PRISM, illustrates both outcomes.

Keywords: Active learning · Evaluation · Oracle bounds · Aleatoric uncertainty · Risk-constrained acquisition

1 Introduction

How much better could an acquisition strategy be? Active-learning research increasingly answers with an *oracle*: a selector that, unlike any deployable strategy, may look at the labels it acquires and at the evaluation set, and so traces the best attainable learning curve. Recent benchmarks construct greedy oracles that add, step by step, the candidate whose *realized* label most improves test performance [43,44], and best-of-strategies oracles that pick, per batch, the strategy whose labelled batch helps most [19]. The gap between oracle and practical strategies is then read as headroom: state-of-the-art strategies are reported to "fall noticeably short of oracle performance, especially in large-scale datasets" [19]. This complements a broader effort to make active-learning evaluation reliable [29,20].

An oracle that selects with realized labels is an *ex-post* benchmark. Part of its advantage is unreachable by construction: when outcomes are random given the covariates, some feasible label set will turn out to be lucky, and only hindsight can find it. The distinction between ex-post and ex-ante regret is familiar in decision-focused learning and robust optimization [35], and in bandits the gap between expected regret and pseudo-regret grows with the noise of the arms [6,2]. It has not been measured for oracle bounds in label acquisition, where it decides whether an "oracle gap" signals room for better strategies or merely outcome noise. Labels in credit, healthcare and marketing record stochastic outcomes rather than near-deterministic categories, so the question matters most where labels are bought with consequential actions.

Contributions.

- **Decomposition and estimator.** We split an oracle's reported lead into test-set reuse, an *aleatoric floor* that no deployable strategy can close, and a *deployable gap*, and estimate all three on real data (Fig. 1) by resampling hidden labels from a probability model fitted on disjoint records. The estimator is validated against known probabilities and calibrated against model-free realized outcomes (Secs. 3, 5).
- **A predictable floor.** The floor is set by label noise between near-tied leading sets: a closed-form bracket contains it in all 87 settings we examined, its share of the oracle gap rises with label noise across eight cohorts as we predicted before running, and under a risk budget it grows with the budget in every cohort (Sec. 6).
- **What remains.** On low-noise benchmark data the oracle's lead is mostly real; on noisy outcomes 58–100% of it is reuse and floor. Who closes the deployable remainder depends on whether a strategy's information proxy matches the learner; DC-PRISM, a conservative policy for risk-constrained acquisition, wins on one domain and loses on another, and a well-powered delayed replay confirms the ordering (Secs. 6, 7).

2 Setting

At round t a learner holds labeled history D_t, the covariates of pending acquisitions O_t (selected but not yet resolved), an unlabeled candidate pool C_t and target covariates Q_t. Candidate i has a visible exposure $e_i \geq 0$ and a binary outcome Y_i revealed only after acquisition. A strategy chooses $S_t \subseteq C_t$ from visible information $\mathcal{H}_t$. Learning is measured by the Brier score [5] of a downstream learner (an ℓ_2-logistic model, gradient boosting, or a one-hidden-layer network); $L(S; Y)$ is its test loss after fitting on $D \cup O \cup S$, where Y collects the hidden training outcomes of O and S.

Risk budget. Acquiring a label is an action with exposure, as when a lender approves an applicant to observe repayment [26,42]. With a risk score q_i, $\ell =$

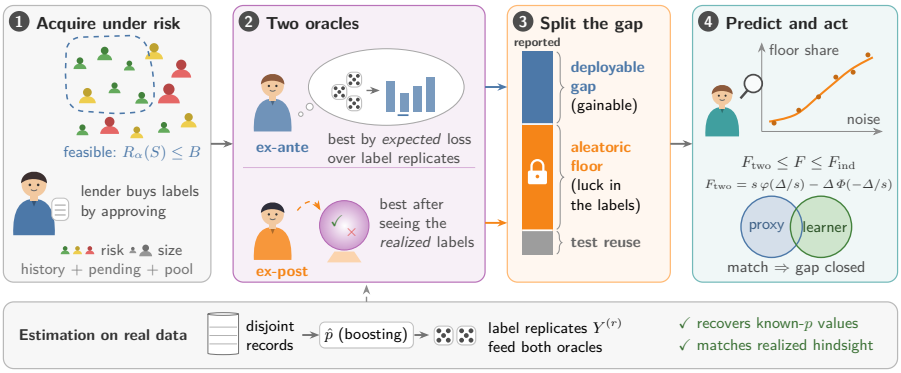


Fig. 1. Overview. (1) In risk-constrained acquisition a budget defines the feasible label sets. (2) An ex-ante oracle picks the set with the best expected loss over label replicates; an ex-post oracle picks the best set after seeing the realized labels, as benchmark oracles do. (3) The reported oracle gap splits into test-set reuse, an aleatoric floor no strategy can reach, and a deployable gap. (4) The floor follows a closed-form bracket and grows with label noise; the deployable gap is closed only by strategies whose information proxy matches the learner. Replicates come from $\hat{p}$ fitted on disjoint records and are checked against known probabilities and realized outcomes. Bar proportions follow the real-label row of Table 1.

$\log(1/\alpha)$ and $\phi(q) = q(1-q)$ for $q \le \frac{1}{2}$, $\frac{1}{4}$ otherwise, feasible sets satisfy

$$R_\alpha(S) = \sum_{i \in S} e_i q_i + \Big(2\ell \sum_{i \in S} e_i^2 \phi(q_i)\Big)^{1/2} + \frac{\ell}{3} \max_{i \in S} e_i \le B_t, \tag{1}$$

a sub-gamma bound [4] under which realized adverse exposure exceeds $R_\alpha(S)$ with probability at most $\alpha = 0.1$ if $p_i \le q_i$. Our scores q_i (a Wilson-shifted [45] high quantile of 40 Laplace draws of the fitted logit) do not guarantee $p_i \le q_i$, so (1) is the operational feasible region shared by all strategies. The budget B_t is 1.03 times the envelope of the k lowest-risk candidates ($k = 6$ unless stated); the *quantity reference* M_K is the largest feasible set, found by enumeration for at most 16 candidates and otherwise by an exposure-stratified search that certified its cardinality in 1,320 of 1,408 states and was off by one label otherwise. For benchmark cohorts without a natural exposure we set $e_i = 1$.

3 Decomposing the Oracle Gap

Family oracle. Fix $\mathcal{H}$ and a family $\mathcal{F}$ of feasible sets built from visible information and containing $M = M_K$. With $\bar{L}$ the test loss with test-label noise integrated out,

$$H_{\text{post}} = \mathbb{E}\big[\bar{L}(M;Y) - \min_{S \in \mathcal{F}} \bar{L}(S;Y) \mid \mathcal{H}\big], \tag{2}$$

$$H_{\text{pre}} = \mathbb{E}[\bar{L}(M;Y) \mid \mathcal{H}] - \min_{S \in \mathcal{F}} \mathbb{E}[\bar{L}(S;Y) \mid \mathcal{H}]. \tag{3}$$

H_{post} is the lead of an oracle that picks the best set after seeing the hidden training labels; H_{pre} is the lead of the best fixed choice. Their difference, the *aleatoric floor* $F = H_{\text{post}} - H_{\text{pre}} \geq 0$ (since $\mathbb{E}\min \leq \min \mathbb{E}$), is unreachable by any deployable strategy; it is the expected value of perfect information about the training outcomes [17]. For a strategy with expected gain Γ_π over M, $H_{\text{post}} - \Gamma_\pi = F + (H_{\text{pre}} - \Gamma_\pi)$, and only the second term, the *selection gap*, can be closed.

Benchmark oracle. Benchmarks use a greedy oracle that adds the feasible candidate whose realized label most reduces loss on the evaluation set, and score it on the same set [43,44]. We split the test labels into halves T_1, T_2 and report three quantities over replicates $Y^{(r)}$ of the training labels: the *reported* gap (oracle selects and is scored on T_1), the *ex-post* gap (same oracle scored on T_2), and the *deployable* gap of an ex-ante greedy oracle that selects by mean T_1 loss over half the replicates and is scored on T_2 over the other half. Then

$$\text{reported} = \underbrace{(\text{reported} - \text{ex-post})}_{\text{test reuse}} + \underbrace{(\text{ex-post} - \text{deployable})}_{\text{aleatoric floor}} + \text{deployable}.$$

The ex-ante oracle is itself greedy, so the deployable gap may be underestimated and the reuse-plus-floor share overestimated; we therefore read that share as an upper bound.

A closed-form bracket for the floor. Let $d_S = \bar{L}(M;Y) - \bar{L}(S;Y)$ with mean μ_S and standard deviation σ_S over replicates. If the two leading sets b, r (by μ) differ by $D = d_b - d_r \sim \mathcal{N}(\Delta, s^2)$, choosing between them with hindsight is worth

$$F_{\text{two}} = s\,\varphi(\Delta/s) - \Delta\,\Phi(-\Delta/s), \tag{4}$$

a lower bound on the family floor, since the oracle also chooses among the other sets. Treating all sets as independent normals, $F_{\text{ind}} = \mathbb{E}\max_S(\mu_S + \sigma_S Z_S) - \max_S \mu_S$ ignores their positive correlation and overstates the floor. When the leaders are near-tied ($\Delta \approx 0$), $F_{\text{two}} \approx s/\sqrt{2\pi}$: the floor scales with the label-noise-induced spread s, which grows with outcome variance $p(1-p)$, with the number of acquired outcomes, and with the learner's sensitivity to individual labels. This is the finite-family analogue of the $\sqrt{n \log K}$ gap between expected regret and pseudo-regret in bandits [2].

Estimation without known probabilities. Given outcome probabilities p, we draw R replicates of the hidden training labels and evaluate every set on all replicates with common random numbers. The plug-in $\widehat{H}_{\text{pre}}$ is biased upward by selecting on noise [37,7] and a two-fold cross-fitted estimate [10] downward, so they bracket H_{pre}. On real data p is unknown; we replace it by $\hat{p}$ from gradient boosting fitted on cohort records disjoint from the state (up to 20,000), and score test loss on the recorded test labels. The recorded training outcomes give one draw of every set's loss, so the realized hindsight gap $\bar{L}(M;y) - \min_S \bar{L}(S;y)$ is model-free and its mean must match the $\hat{p}$-based H_{post}.

4 Strategies under Evaluation

All strategies share the state, q, the budget and (1). *Safe-MinRisk* returns M_K; *fixed count* takes a fixed fraction of K by point value; *Fisher-cost* packs by single-label variance reduction per exposure. Four rules pack greedily until no feasible candidate remains: *Random*, *Uncertainty* [28], *BAIT-greedy* on $\mathrm{tr}(F_Q(H_{D+O} + A_S)^{-1})$ [1], and *TargetVar-greedy* on the target-weighted proxy

$$V_\theta(S) = \mathrm{tr}\left[W_Q\{H_{D+O}^{-1} - (H_{D+O} + A_S)^{-1}\}\right], \tag{5}$$

where H_{D+O} is the logistic Fisher information of history and pending covariates, A_S that of S, and W_Q weights target directions by their Brier sensitivity [48,3]. *Proxy-argmax* picks $\arg\max V_{\hat\theta}$ over the evaluation family. *DC-PRISM* is a conservative policy designed for risk-constrained acquisition with pending outcomes: it evaluates (5) on a family anchored at M_K under 20 Laplace parameter draws [30], scores each set by the 0.2-quantile of $\tanh\{(V_m(S) - V_m(M_K))/2.5V_m(M_K)\}$ minus a quantity penalty $0.22(K - |S|)_+/K$, and returns M_K unless the best score exceeds 0.08. A selection takes about 50 ms with 24 candidates. All information proxies here are derived from the logistic model.

5 Validating the Protocol

Studies. Every study fixed its states, seeds, comparison families and a material-effect threshold of 5×10^{-4} Brier before running, with protocol and code frozen in version control; we report paired means with 95% Bonferroni simultaneous intervals and Holm-adjusted tests [16]. All values are in 10^{-3} Brier. Cohorts: four credit cohorts (HMEQ, LendingClub, PPDai, UCI Credit [33,27,13,14,47]), 30-day hospital readmission [38] with days in hospital as exposure, bank-marketing subscription, Spambase, and USPS digits (≥ 5 versus < 5), the last as raw pixels and as 10 principal components. Each state is the first round of a delayed replay with 68 historical labels, 14 pending and 24 candidate records; candidates are drawn at random (*natural* stream) or, for credit and readmission, also from an *engineered* stream that ranks covariates into low-risk, pending-aligned, complementary and tail candidates. Controlled worlds and *semi-synthetic* states (real covariates, outcomes from a known logit) provide known p.

Estimated probabilities recover known-probability values. On 512 semi-synthetic states, boosting $\hat p$ fitted on auxiliary records labeled from the true p gives estimates whose bias against the known-p protocol lies within ± 0.5 in 95 of 96 checks (8 groups $\times$ 12 quantities); mean biases of H_{post}, H_{pre}, the floor and the main strategy gains are at most 0.20, with state-level correlations of 0.73–0.95 for H_{pre}. A logistic $\hat p$ underestimates H_{pre} by 0.26–0.28 where the truth is nonlinear, so the estimator needs a flexible $\hat p$.

Table 1. Benchmark greedy oracle under the risk budget, logistic learner ($\times 10^{-3}$ Brier). Reported gap = test reuse + aleatoric floor + deployable gap. Last column: lead over the best fixed practical strategy, as reported → after removing reuse and floor.

Setting	States	Reported	Reuse	Floor	Deployable	Lead over best
Controlled worlds	64	29.5	4.5	3.5	21.5	15.8→7.8
Semi-synthetic (known p)	256	7.2	2.9	3.9	0.3	6.9→0.0
Credit, engineered stream	128	8.0	0.6	4.1	3.3	6.7→2.0
Credit, natural stream	128	3.4	0.8	1.7	0.9	3.2→0.6
Readmission, both streams	64	2.3	0.3	1.4	0.5	2.0→0.3

The estimator is calibrated on recorded outcomes. With the logistic and boosting learners, the $\hat{p}$-based H_{post} agrees with the mean model-free realized gap in all 16 cohort–learner cells of the eight cohorts and in all 8 credit cells of an independent 128-seed batch (every 95% interval contains zero). With the neural learner it agrees in 8 of 9 cohorts; on PPDai it overstates the realized gap (7.4 against 5.0, difference 2.4 $[0.6, 4.2]$), a limit of the substitution that we flag in the results.

The decomposition does not depend on the family. Enlarging each family with 64 random feasible sets (from 22–27 to 81–88 sets) leaves the deployable share $H_{\text{pre}}/H_{\text{post}}$ essentially unchanged: 80–83% in controlled worlds at every subfamily size from 2 to 80, and 6–38% on semi-synthetic arms, where it rises slightly rather than falls with $|\mathcal{F}|$.

6 Findings

Most of a benchmark oracle's lead on noisy outcomes is reuse and floor. Table 1 applies the benchmark oracle to settings with known p and to real-label credit and readmission states (logistic learner). Where p is known and the oracle is genuinely better (controlled worlds), its lead over the best fixed strategy halves after correction (15.8 → 7.8). On semi-synthetic states with real covariates it vanishes (6.9 → 0.0): reuse (2.9) and floor (3.9) make up 95% of the reported gap. On recorded outcomes the floor alone is 52% of the reported gap and the oracle's lead over the best fixed strategy shrinks from 4.5 to 1.3 $[0.9, 1.6]$. The floor is material in 18 of 19 groups.

The floor's share rises with label noise. Before running the eight-cohort study we registered that the floor's share of the oracle gap would be smaller on the two cohorts with the lowest noise index (mean $\hat{p}(1-\hat{p})$ over candidates) and would increase with that index. Both held (Table 2): the lowest-noise cohorts, Spambase and USPS, had a floor share of 14% against 59% for the other six, and the Spearman correlation between noise index and floor share was 0.83 (logistic learner) and 0.67 (neural learner). On USPS with raw pixels and a neural learner, the setting of a recent benchmark [44], the floor is 10% of the oracle gap and the oracle's lead remains 8.0 after correction: the benchmark's

Table 2. Benchmark oracle along a label-noise gradient (natural stream, 48 states per cohort with the logistic and 32 with the neural learner; $\times 10^{-3}$ Brier). Noise: mean $\hat{p}(1-\hat{p})$. Reuse and floor: % of the reported gap. Lead: over the best fixed strategy, reported → deployable.

		Logistic learner				Neural learner (MLP)			
Cohort	Noise	Gap	Reuse	Floor	Lead	Gap	Reuse	Floor	Lead
USPS, raw pixels	0.036			–		18.7	32	10	15.9→8.0
Spambase	0.054	7.5	32	16	5.1→1.5	14.1	25	29	11.4→3.8
USPS, 10 PCs	0.065	10.8	12	13	7.7→5.0	16.6	23	29	15.0→6.4
Readmission	0.081	1.3	29	66	1.1→−0.1	4.8	23	54	4.0→0.3
Bank Marketing	0.091	4.5	15	39	4.1→1.7	11.9	34	29	10.3→2.8
HMEQ	0.101	3.4	15	42	3.0→1.0	7.8	26	39	7.2→2.1
PPDai	0.108	2.8	17	60	2.7→0.5	10.4	20	38	9.1→3.0
UCI Credit	0.136	3.4	15	67	3.2→0.4	8.2	47	38	7.8→0.8
LendingClub	0.144	2.7	8	79	2.8→0.4	8.5	39	53	8.6→0.7

message that strategies leave room is largely sound there. On credit, readmission and marketing outcomes, 58–100% (logistic) and 67–93% (neural) of the lead is reuse and floor. Test reuse is a separate, noise-independent bias: it is up to a third of the reported gap on Spambase and larger with the neural learner.

The floor is predictable. Across all studies, the measured family floor lies between the two-leader lower bound (4) and the independent-normal upper bound in all 87 cohort–learner–budget settings (Fig. 2a). The lower bound is tight up to a near-constant factor: the measured floor is about 2.7 times (4) with $R^2 = 0.92$ (logistic) and 0.97 (boosting) across groups, and state-level Spearman correlations of 0.68 and 0.85. The floor depends on the learner, as the bracket predicts through s: with the same states and labels it spans 0.7–4.4 for the logistic learner and 0.9–8.3 for boosting, and the benchmark-oracle floor is 0.8–2.3 for the logistic and 1.8–4.9 for the neural learner. Whereas the deployable gap varies by two orders of magnitude across settings, the floor stays within one, so hindsight comparisons mislead most where true differences are small. Fig. 2b shows the registered noise ordering for both learners.

A risk budget raises the floor. The bracket implies that admitting more random outcomes raises the floor. We registered that the floor would increase with the risk budget and varied k in B_t from 3 to 6 to 12 (Fig. 3). The floor rose monotonically in all eight cohorts, from 0.32–0.87 at $k = 3$ to 0.91–1.90 at $k = 12$ (pooled paired increase 0.74 $[0.67, 0.82]$), and the bracket held at every budget. The mean outcome variance of the reference set changed by at most 0.02 within a cohort, so the increase comes from acquiring more random outcomes rather than riskier ones. In risk-constrained acquisition the budget therefore controls how far an oracle comparison can be trusted.

Who closes the deployable gap depends on the learner. Table 3 reports, on the same real-label states, the deployable gap and the gains of three strategies under two learners. With the logistic learner, information proxies derived from that

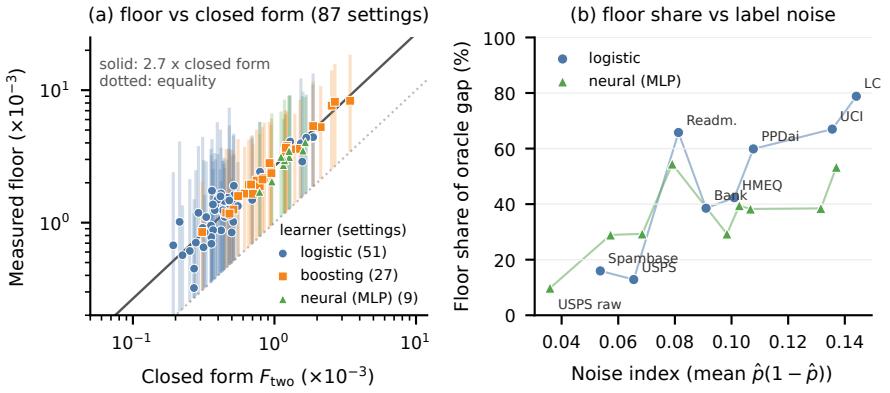


Fig. 2. (a) Measured family floor against the two-leader closed form (4) in all 87 cohort–learner–budget settings; vertical bars span the bracket $[F_{\text{two}}, F_{\text{ind}}]$, which contains every measured value. The solid line is the median ratio 2.7 ($R^2 = 0.89$ on the log scale). (b) Floor share of the benchmark oracle's reported gap against the label-noise index, logistic (48 states per cohort) and neural learner (32).

model close a large part of it where it is largest: on engineered HMEQ and LendingClub TargetVar-greedy and Proxy-argmax capture 1.5–2.1 of 3.2–3.4 (in an independent 128-seed batch, 2.5–2.9 of 4.9–5.7, while DC-PRISM trails TargetVar-greedy by 2.8 $[0.0, 5.6]$ on LendingClub). With the boosting learner the deployable gap is still 0.7–2.7, DC-PRISM's selection gap is material in six of eight credit cells, and no strategy captures more than 0.65: proxies that match the logistic model do not transfer to the boosting learner. On readmission the deployable gap is small (0.7–0.9) and no strategy differs decisively.

DC-PRISM in controlled worlds. In worlds built around directional pending information, 83% of hindsight headroom is deployable and DC-PRISM captures two thirds of it: its gain over Safe-MinRisk is 12.2 $[7.6, 16.8]$ with 3.1 instead of 6 labels, and remains material in all twelve generator variants we tried (10.8–31.4; pending overlap, risk–information coupling, no misspecified tail, 48 candidates with $K = 12$). A common-family ablation attributes this gain to the proxy evaluated over an anchored family (Proxy-argmax alone gains 13.1 $[6.1, 20.2]$); the conservative layer adds an unconfirmed 3.0 $[-0.2, 6.2]$, and pending-awareness is not isolated by our evidence.

7 Outcome Check: Delayed Replay

The protocol scores single rounds. To check its ordering under dynamics we replayed ten rounds (exploration labels after one round, ordinary labels after two; ALC-Brier averages ten round-start scores) with 512 fresh seeds per credit cell

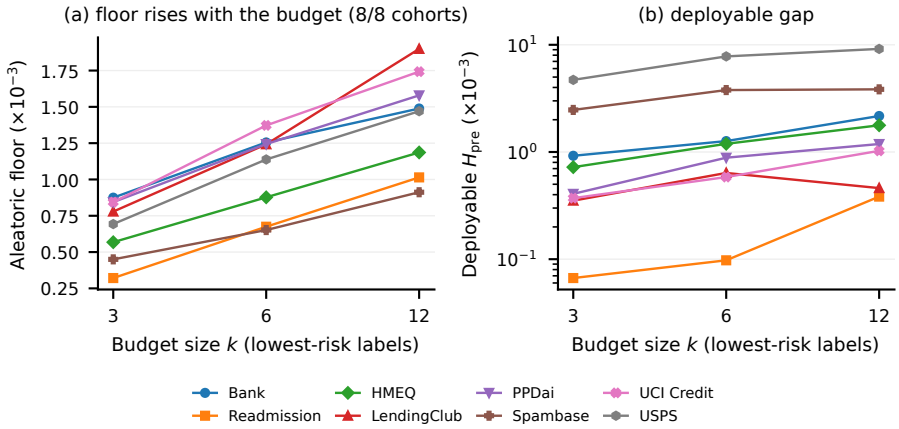


Fig. 3. Risk budget as a lever (logistic learner, 32 states per cohort and budget). The budget is sized for the k lowest-risk candidates. (a) The aleatoric floor rises monotonically in all eight cohorts. (b) The deployable gap (log scale) moves much less uniformly.

and 256 per readmission stream, numbers registered from the variance of an earlier batch. Table 4 shows the ordering the protocol predicts. Where information-greedy rules captured the deployable gap and DC-PRISM did not (engineered HMEQ and LendingClub), DC-PRISM is materially worse than Fisher-cost, BAIT-greedy and TargetVar-greedy (2.5–3.8); where its gain over the quantity reference was significant, it beats Safe-MinRisk and fixed count (2.2–3.1); on engineered readmission it beats BAIT-greedy and TargetVar-greedy (1.3, 1.4); and where little deployable gap exists (natural streams, engineered UCI) no comparison is material. Realized adverse exposure exceeded the envelope in 0–4 of 40,960 acquisition rounds per credit policy and in none of 5,120 for readmission.

8 Related Work

Evaluating active learning. Recent work shows that active-learning gains are fragile under realistic evaluation [29], sensitive to uncontrolled randomness [20] and inconsistent across domains [43,44], and uses oracles to bound attainable performance [44,19]. Werner et al. note that their greedy oracle overfits the test set [43]. We quantify that reuse and add an aleatoric component that remains after it is removed.

Regret notions and uncertainty. Ex-ante and ex-post regret are distinguished in decision-focused learning and robust optimization [35]; expected regret exceeds pseudo-regret by a noise term in bandits [6,2]; proper scoring rules such as the Brier score separate aleatoric from epistemic uncertainty [15]; and selecting the best of noisy estimates is optimistic [37,7], the value of perfect information being

Table 3. Deployable gap H_{pre} (cross-fitted) and gains over M_K on real-label states (64 credit and 128 readmission states per cell; $\times 10^{-3}$ Brier). DC: DC-PRISM; TV: TargetVar-greedy; Proxy: Proxy-argmax.

	Logistic learner				Boosting learner			
State	H_{pre}	DC	TV	Proxy	H_{pre}	DC	TV	Proxy
HMEQ, eng.	3.21	0.09	1.51	1.64	2.32	0.48	0.15	0.44
LC, eng.	3.37	1.15	2.01	2.12	1.53	0.05	−0.42	−0.19
PPDai, eng.	1.87	0.37	0.04	0.07	1.18	0.33	0.17	0.11
UCI, eng.	0.16	−0.08	−0.08	−0.08	0.84	0.36	0.42	0.34
HMEQ, nat.	1.26	−0.22	0.21	0.12	2.66	0.24	0.64	0.65
LC, nat.	0.91	−0.04	−0.20	−0.05	1.45	−0.07	0.04	−0.04
PPDai, nat.	0.59	0.10	−0.17	0.02	1.17	−0.03	−0.18	−0.10
UCI, nat.	0.92	0.07	0.19	0.32	1.60	0.03	0.13	0.08
Readm., eng.	0.84	0.09	−0.14	−0.34	0.92	−0.01	−0.25	−0.28
Readm., nat.	0.67	0.06	0.30	0.36	0.72	0.00	−0.10	−0.07

Table 4. Delayed replay: comparator minus DC-PRISM ALC-Brier ($\times 10^{-3}$; positive favors DC-PRISM; bold: material under the registered family).

	Credit, engineered				Credit, natural				Readmission	
Comparator	HMEQ	LC	PPDai	UCI	HMEQ	LC	PPDai	UCI	eng.	nat.
Safe-MinRisk	**2.4**	0.4	**3.1**	−0.1	0.2	0.2	0.4	0.1	0.0	0.2
Fixed count	**2.2**	0.4	**2.4**	−0.1	0.1	0.1	0.3	0.0	0.4	0.2
Random	0.6	−0.3	**2.4**	0.0	0.3	0.5	0.6	0.4	0.8	0.5
Uncertainty	0.5	1.4	0.7	0.0	0.4	0.4	0.1	0.1	0.4	0.0
Fisher-cost	**−3.8**	**−3.3**	0.8	−0.1	−0.3	−0.1	0.1	−0.1	1.0	0.0
BAIT-greedy	**−3.6**	**−2.7**	0.3	0.1	−0.2	0.0	0.1	0.0	**1.3**	0.1
TargetVar-greedy	**−3.5**	**−2.5**	0.0	0.0	−0.2	−0.1	0.0	−0.1	**1.4**	0.0

classical [17,32]. We bring these ideas to oracle bounds for label acquisition and make the aleatoric part measurable and predictable on real data.

Acquisition under risk and delay. Acquisition criteria range from uncertainty and transductive design to Fisher information, mutual and predictive information and decision loss [11,36,28,48,1,24,3,18,8]; budgets and knapsack constraints are handled by greedy selection [25,40,41]. Selective labels [26,42,23,22,12], safe exploration and safe active learning [39,34,46] and delayed feedback [21,9,31] motivate the risk-constrained setting in which we study the floor.

9 Discussion and Conclusion

An oracle's lead is a natural yardstick for how much an acquisition strategy could improve. Our decomposition shows that the yardstick carries an aleatoric floor, set by label noise among near-tied leading sets, together with test reuse. The floor is predictable in closed form, rises with outcome noise and with the risk budget, and depends on the learner. On low-noise benchmark data the oracle's lead is mostly real; on noisy outcomes most of it is not, and what remains is captured only by strategies whose information proxy matches the learner. We recommend reporting oracle gaps on held-out evaluation data, together with

the floor bracket (4) or the decomposition when outcome probabilities can be estimated from disjoint records.

Limitations. Learners and pools are small (logistic, small boosting and one-hidden-layer networks; 24 candidates, 6–256 features); large pretrained models were not studied. The deployable gap is measured with a greedy ex-ante oracle, so the reuse-plus-floor share is an upper bound. The real-data estimator needs a flexible $\hat{p}$ from disjoint records; it failed calibration once (PPDai, neural learner), and in lending those records contain only approved applicants [12]. Multiclass benchmarks were binarized, splits are random rather than temporal, BAIT-greedy and TargetVar-greedy are adaptations, and the envelope's guarantee is not implied by our risk scores, although realized exceedance stayed far below α. Code, frozen protocols with commit hashes, aggregate results and verifiers are available in an anonymized repository.

References

1. Ash, J., Goel, S., Krishnamurthy, A., Kakade, S.: Gone fishing: Neural active learning with Fisher embeddings. In: Advances in Neural Information Processing Systems. vol. 34, pp. 8927–8939 (2021), https://proceedings.neurips.cc/paper_files/paper/2021/hash/4afe044911ed2c247005912512ace23b-Abstract.html
2. Audibert, J.Y., Bubeck, S.: Regret bounds and minimax policies under partial monitoring. Journal of Machine Learning Research **11**, 2785–2836 (2010)
3. Bickford Smith, F., Kirsch, A., Farquhar, S., Gal, Y., Foster, A., Rainforth, T.: Prediction-oriented Bayesian active learning. In: Proceedings of the 26th International Conference on Artificial Intelligence and Statistics. Proceedings of Machine Learning Research, vol. 206, pp. 7331–7348 (2023), https://proceedings.mlr.press/v206/bickfordsmith23a.html
4. Boucheron, S., Lugosi, G., Massart, P.: Concentration Inequalities: A Nonasymptotic Theory of Independence. Oxford University Press (2013). https://doi.org/10.1093/acprof:oso/9780199535255.001.0001
5. Brier, G.W.: Verification of forecasts expressed in terms of probability. Monthly Weather Review **78**(1), 1–3 (1950). https://doi.org/10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2
6. Bubeck, S., Cesa-Bianchi, N.: Regret analysis of stochastic and nonstochastic multi-armed bandit problems. Foundations and Trends in Machine Learning **5**(1), 1–122 (2012)
7. Cawley, G.C., Talbot, N.L.C.: On over-fitting in model selection and subsequent selection bias in performance evaluation. Journal of Machine Learning Research **11**, 2079–2107 (2010)
8. Chaloner, K., Verdinelli, I.: Bayesian experimental design: A review. Statistical Science **10**(3), 273–304 (1995)
9. Chapelle, O.: Modeling delayed feedback in display advertising. In: Proceedings of the 20th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. pp. 1097–1105 (2014)
10. Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., Robins, J.: Double/debiased machine learning for treatment and structural parameters. The Econometrics Journal **21**(1), C1–C68 (2018)

11. Cohn, D.A., Ghahramani, Z., Jordan, M.I.: Active learning with statistical models. Journal of Artificial Intelligence Research **4**, 129–145 (1996). https://doi.org/10.1613/jair.295
12. Crook, J., Banasik, J.: Does reject inference really improve the performance of application scoring models? Journal of Banking & Finance **28**(4), 857–874 (2004)
13. He, H.: PPDaiData. figshare dataset, CC BY 4.0 (2021). https://doi.org/10.6084/m9.figshare.13573856.v1, 55,596 loans from the PaiPaiDai P2P lending platform
14. He, H., Zhang, W., Zhang, S.: A novel ensemble method for credit scoring: Adaption of different imbalance ratios. Expert Systems with Applications **98**, 105–117 (2018). https://doi.org/10.1016/j.eswa.2018.01.012
15. Hofman, P., Sale, Y., Hüllermeier, E.: Quantifying aleatoric and epistemic uncertainty with proper scoring rules. arXiv:2404.12215 (2024)
16. Holm, S.: A simple sequentially rejective multiple test procedure. Scandinavian Journal of Statistics **6**, 65–70 (1979). https://doi.org/10.2307/4615733
17. Howard, R.A.: Information value theory. IEEE Transactions on Systems Science and Cybernetics **2**(1), 22–26 (1966)
18. Huang, Z., Bickford Smith, F., Rainforth, T.: Loss-driven Bayesian active learning. In: Proceedings of the 29th International Conference on Artificial Intelligence and Statistics. Proceedings of Machine Learning Research, vol. 300, pp. 5140–5148 (2026), https://proceedings.mlr.press/v300/huang26b.html
19. Huseljic, D., Hahn, P., Herde, M., Sandrock, C., Sick, B.: BoSS: A best-of-strategies selector as an oracle for deep active learning. Transactions on Machine Learning Research (2026), arXiv:2603.13109
20. Ji, Y., Kaestner, D., Wirth, O., Wressnegger, C.: Randomness is the root of all evil: More reliable evaluation of deep active learning. In: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV). pp. 3943–3952 (2023)
21. Joulani, P., György, A., Szepesvári, C.: Online learning under delayed feedback. In: Proceedings of the 30th International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 28, pp. 1453–1461 (2013)
22. Keswani, V., Mehrotra, A., Celis, L.E.: Fair classification with partial feedback: An exploration-based data collection approach. In: Proceedings of the 41st International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 235, pp. 23547–23576 (2024), https://proceedings.mlr.press/v235/keswani24a.html
23. Kilbertus, N., Gomez Rodriguez, M., Schölkopf, B., Muandet, K., Valera, I.: Fair decisions despite imperfect predictions. In: Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics. Proceedings of Machine Learning Research, vol. 108, pp. 277–287 (2020)
24. Kirsch, A., van Amersfoort, J., Gal, Y.: BatchBALD: Efficient and diverse batch acquisition for deep Bayesian active learning. arXiv:1906.08158 (2019). https://doi.org/10.48550/arXiv.1906.08158
25. Krause, A., Guestrin, C.: Near-optimal nonmyopic value of information in graphical models. In: Proceedings of the 21st Conference on Uncertainty in Artificial Intelligence (UAI). pp. 324–331 (2005)
26. Lakkaraju, H., Kleinberg, J., Leskovec, J., Ludwig, J., Mullainathan, S.: The selective labels problem: Evaluating algorithmic predictions in the presence of unobservables. In: Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. pp. 275–284 (2017). https://doi.org/10.1145/3097983.3098066

27. LendingClub Corporation: LendingClub historical loan data (accepted loans), originally published by LendingClub; processed copy obtained from the DePaul University Economics Data Archive, https://bigblue.depaul.edu/jlee141/econdata/LendingClub_LoanData/, accessed 4 November 2026
28. Lewis, D.D., Gale, W.A.: A sequential algorithm for training text classifiers. In: Proceedings of the 17th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval. pp. 3–12 (1994)
29. Lüth, C.T., Bungert, T.J., Klein, L., Jaeger, P.F.: Navigating the pitfalls of active learning evaluation: A systematic framework for meaningful performance assessment. In: Advances in Neural Information Processing Systems. vol. 36, pp. 9789–9836 (2023)
30. MacKay, D.J.C.: A practical Bayesian framework for backpropagation networks. Neural Computation **4**(3), 448–472 (1992). https://doi.org/10.1162/neco.1992.4.3.448
31. Pham, T., Kottke, D., Krempl, G., Sick, B.: Stream-based active learning for sliding windows under the influence of verification latency. Machine Learning **111**(6), 2011–2036 (2022). https://doi.org/10.1007/s10994-021-06099-z
32. Raiffa, H., Schlaifer, R.: Applied Statistical Decision Theory. Harvard University Press, Boston (1961)
33. SAS Institute: Home equity data set. SAS example datasets, https://support.sas.com/documentation/onlinedoc/viya/exampledatasets/home_equity.csv, accessed 4 November 2026
34. Schreiter, J., Nguyen-Tuong, D., Eberts, M., Bischoff, B., Markert, H., Toussaint, M.: Safe exploration for active learning with Gaussian processes. In: Machine Learning and Knowledge Discovery in Databases (ECML PKDD). LNCS, vol. 9286, pp. 133–149. Springer (2015)
35. Schutte, N., Postek, K., Yorke-Smith, N.: Robust losses for decision-focused learning. arXiv:2310.04328 (2023)
36. Settles, B.: Active learning literature survey. Computer Sciences Technical Report 1648, University of Wisconsin–Madison (2009)
37. Smith, J.E., Winkler, R.L.: The optimizer's curse: Skepticism and postdecision surprise in decision analysis. Management Science **52**(3), 311–322 (2006)
38. Strack, B., DeShazo, J.P., Gennings, C., Olmo, J.L., Ventura, S., Cios, K.J., Clore, J.N.: Impact of HbA1c measurement on hospital readmission rates: Analysis of 70,000 clinical database patient records. BioMed Research International **2014**, 781670 (2014). https://doi.org/10.1155/2014/781670
39. Sui, Y., Gotovos, A., Burdick, J.W., Krause, A.: Safe exploration for optimization with Gaussian processes. In: Proceedings of the 32nd International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 37, pp. 997–1005 (2015)
40. Sviridenko, M.: A note on maximizing a submodular set function subject to a knapsack constraint. Operations Research Letters **32**(1), 41–43 (2004)
41. Tomanek, K., Hahn, U.: A comparison of models for cost-sensitive active learning. In: Coling 2010: Posters. pp. 1247–1255 (2010), https://aclanthology.org/C10-2143/
42. Wei, D.: Decision-making under selective labels: Optimal finite-domain policies and beyond. In: Proceedings of the 38th International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 139, pp. 11035–11046 (2021), https://proceedings.mlr.press/v139/wei21a.html
43. Werner, T., Burchert, J., Schmidt-Thieme, L.: Towards comparable active learning. arXiv:2311.18356 (2023)

44. Werner, T., Burchert, J., Stubbemann, M., Schmidt-Thieme, L.: A cross-domain benchmark for active learning. arXiv:2408.00426 (2024)
45. Wilson, E.B.: Probable inference, the law of succession, and statistical inference. Journal of the American Statistical Association **22**(158), 209–212 (1927). https://doi.org/10.1080/01621459.1927.10502953
46. Wu, Y., Shariff, R., Lattimore, T., Szepesvári, C.: Conservative bandits. In: Proceedings of the 33rd International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 48, pp. 1254–1262 (2016)
47. Yeh, I.C., Lien, C.h.: The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients. Expert Systems with Applications **36**(2), 2473–2480 (2009). https://doi.org/10.1016/j.eswa.2007.12.020
48. Yu, K., Bi, J., Tresp, V.: Active learning via transductive experimental design. In: Proceedings of the 23rd International Conference on Machine Learning. pp. 1081–1088 (2006). https://doi.org/10.1145/1143844.1143980